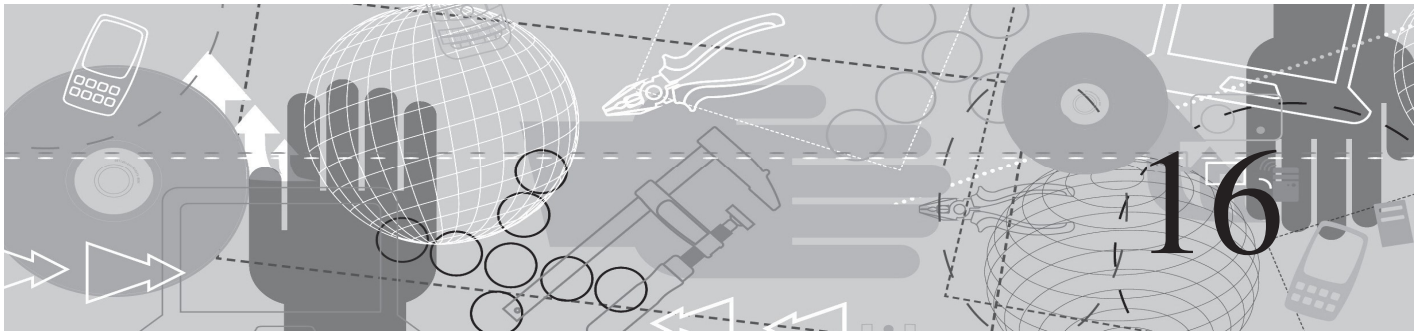


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Fordism and the Golden Age of Atlantic Capitalism

Matt Vidal

INTRODUCTION

The term fordism goes back to the time of Henry Ford. In 1916 the Ford Motor Company produced half a million Model T cars, and by 1921 it had 55 percent of the US market. Ford built on previous American innovations through the use of standardized parts and machinery, which were pioneered in the arms industry under the direction and funding of the US government. Such technologies were further developed by consumer durable goods companies including Singer (sewing machines), Pope Manufacturing (bicycles) and Western Wheel Works (bicycles). While the latter companies marketed their products in the top price category, Ford designed the Model T as a 'car for the masses', being the first to pursue a high-volume, low-cost strategy (Hounshell 1984). He and his engineers developed the concept of flow production, including but not limited to the moving assembly line, first installed in 1913. Due to the extremely demanding environment of machine-paced, flow production, Ford's turnover rate skyrocketed to 380 percent in 1913. As a result, Ford introduced the five-dollar day in 1914, effectively doubling the wage of his workers.

The fordist system was truly revolutionary, not for its contributions to productivity, which are better understood as part of a longer term evolutionary process in which Ford was one of many key players, but for its embrace of the mass market. As Hounshell (1984: 9–10) explains, 'Before the era of the Model T, the word *masses* had carried a largely negative connotation' but 'Ford recognized "the masses" as a legitimate and seemingly unlimited market for the most sophisticated consumer durable product of the early twentieth century'. Indeed, the term mass production was introduced only in 1925, in an article on the topic commissioned by the *Encyclopedia Britannica* and also published in the *New York Times*, ghost-written under Henry Ford's name by one of his spokesmen. Thereafter, mass production became a widely used term,

but fordism retained a specific meaning of high-volume, low-cost, high-wage production of standardized products.

As an academic concept, Antonio Gramsci (1999 [1929–1935]) argued that fordism, with its high wage for semi-skilled workers, was a form of material class compromise in the workplace. This notion of fordism as an institutional basis for incorporating the working class into the capitalist economy was later extended from the workplace to the macroeconomy by Michel Aglietta (2000 [1970]). Fordism was not just a production system but a regime of accumulation, or national growth regime, characterized by mass production combined with institutions supporting mass consumption, including a system of collective bargaining which generalized the class compromise of relatively high and growing wages in return for labor peace. Following the work of Gramsci, Aglietta and a few other seminal contributions (Boyer 1979; Palloix 1976), the concept of fordism has been deployed within a range of disciplines, including business history (Hounshell 1984; Tolliday and Zeitlin 1991), the sociology of work (Gilbert et al. 1992; Vallas 1999; Vidal 2007), industrial relations (Dohse et al. 1985; Tolliday and Zeitlin 1992a), heterodox macroeconomics (Boyer 2000; Lipietz 1988), comparative political economy (Amin, 1995; Peck and Tickell 2000; Piore and Sabel 1984; Sabel 1982; Tickell and Peck 1992) and state theory (Bonefeld and Holloway 1991; Jessop 2002).

The concept of fordism has also received its fair share of criticism. In this chapter I carefully consider such criticism and submit that the concept of fordism is essential for an adequate understanding of the developmental trajectory of twentieth-century capitalism in Europe and North America. The strength of the concept of fordism lies in its multilevel character as a conceptual framework for establishing causal linkages between the micro-organizational and macro-institutional constitution of economies (Aglietta 2000 [1970]; Boyer 1988). The American economy became the most powerful economy in the world over the twentieth century because its fordist *model of production* was embedded within a *national growth regime*, including a Keynesian welfare state and an explicit compromise between big business and big unions, and an *international growth regime* in which the US was the hegemonic state within the Bretton Woods monetary system.

This chapter proceeds as follows. The first section briefly outlines a regulation theoretic approach to fordism as a multilevel analytical framework. The next three sections focus on fordism as a production model in three countries: *classical fordism* in the US, *reluctant fordism* in the UK and *flexible fordism* in Germany. In the section on the US, I deal with the two strongest critiques of the concept of fordism (Clarke 1992; Williams et al. 1992). Ford's Model T production system, based on special-purpose machinery and flow production, was developed by other American firms and generalized into the model of classical fordism, based on general-purpose machinery and batch production. In the UK there was substantial resistance to fordist production due to an enduring institutional legacy from its much longer history of industrialization, while Germany adopted the model more fully, with substantial help from the Marshall Planners, diverging only in cases where it made advances on fordist production in terms of flexibility. The fifth section discusses the international regime of *Atlantic fordism*, based in the Bretton Woods monetary system (1945–1973). The sixth section discusses the leading national growth regimes that were established within Atlantic fordism: *liberal fordism* in the US, *blocked fordism* in the UK and *nonliberal fordism* in Germany. The final section concludes.

THEORY: FORDISM FROM PRODUCTION MODEL TO INTERNATIONAL GROWTH REGIME

Aglietta (2000 [1970]) developed a regulation theory of American capitalism to explain how

class compromise and economic stability were established through institutional forms of regulation based in and around fordist production. Following Boyer (1988), the analysis here distinguishes between three institutional domains: the production model (work and employment relations), the national regime (forms of competition and the state) and the international regime. Here I provide three brief theoretical comments. First, Marx (1990 [1867]) predicted a tendency toward the centralization capital – the growth of large oligopolies through mergers and acquisitions – as a normal outcome of capitalist competition. Nearly a hundred years later, oligopoly was a basic fact of industrialization according to the dean of business history, Alfred Chandler (1962: 25, 30): 'Expanded output often led to overproduction and then to combination [i.e. horizontal mergers]. … The threat of excess capacity appears to have been a primary stimulus to initial combinations in most American industries'. In this chapter I emphasize how oligopoly in the core of the national fordist growth regimes – in conjunction with low levels of international trade under the international Atlantic fordist regime – allowed wages to be taken out of competition.

Second, the relation of national growth regimes to the international regime can be conceptualized through Trotsky's notion of combined and uneven development. 'The privilege of historical backwardness', wrote Trotsky (2001 [1932]: 27) 'permits, or rather compels, the adoption of whatever is ready in advance of any specified date, skipping a whole series of intermediate stages … The fact that Germany and the United States have now economically outstripped England was made possible by the very backwardness of their capitalist development'. Veblen (2003 [1915]: 42, 49, 53) offered a similar analysis in his study of German industrialization. 'Frames of mind' that were 'habituated' to the early phase of the Industrial Revolution, based in a handicraft 'conception' of industry, acted to restrain British technological growth in later phases. Due to the durability of these forms of habituation and their lag in catching up with technical change, the British paid a 'penalty for having been thrown into the lead and so having shown the way'.

Gerschenkron (1962) made the notion of backwardness central to his analysis of German industrialization. In Peck and Theodore's (2007: 760–1) regulation theory of variegated capitalism, combined and uneven development offers a way of 'coming to terms with the causes and forms of capitalism's dynamic polymorphism', including 'combined, multiscalar hybrid' organizational and institutional forms. This theory does not assume that national economies must be institutionally homogeneous or coherent (Vidal 2014). Likewise, Djelic (1998: 2, 9) argued that 'national systems of industrial production did not evolve in a discrete or independent fashion … the partial convergence of national systems of industrial production after 1945 cannot be understood outside of a peculiar geopolitical context'.

Finally, in the introduction to their volume of major contributions to the analysis of fordism as a model of production, Beynon and Nichols (2006a: xviii) argued for the need to move beyond the oversimplified 'binary histories' of fordism and post-fordism. Regulation theory provides a methodological and conceptual apparatus for theorizing how national growth regimes within an international regime may follow similar developmental trajectories yet differ in their concrete institutionalization (for an overview of regulation theory and in-depth elaboration of my approach, see Vidal (2014)). The analysis presented here is realist, rather than ideal typical, meaning that it seeks to make claims about actually existing regimes at each level: organizational, sectoral, national and international. Whether under the name of fordism, mass production, or the American model, classical fordism was a reality-derived production model that was explicitly diffused to Western Europe (Djelic 1998). Beynon and Nichols (2006a: xiv) also note that using the

fordism/post-fordism dichotomy as a frame for comparative and historical analysis 'puts a very high premium on having specified fordism as a method of production correctly in the first place'. Following my in-depth historical analysis, I thus carefully summarize each regime in a series of tables listing key characteristics, subject to refinement based on new evidence.

THE CLASSICAL FORDIST PRODUCTION MODEL IN THE USA

Precursors to Classical Fordism: From the 'American System of Manufactures' to the Model T System

In the nineteenth century a distinctly 'American system of manufactures' emerged: special-purpose machine tools were arranged in a sequential operation to produce interchangeable parts. It was called the 'American system' by Europeans who were still using general-purpose machine tools run by skilled machinists producing custom parts. According to Hounshell (1984), the original ideas for standardized parts were developed by a French military general, Jean-Baptiste de Gribeauval, in the 1760s. The French military's experiments with interchangeable parts caught the attention of Thomas Jefferson who lobbied for the US Congress to take up the idea. From the 1790s to the 1820s, the War Department funded the Federal Armory at Springfield and private contractors, most notably Simeon North and Eli Whitney, to develop standardized arms. This goal was not realized until the American engineer Thomas Blanchard, building on ideas borrowed from the French engineer Marc I. Brunel, designed a sequence of special-purpose machines to build standardized rifle stocks. By 1826 Blanchard had perfected his sequential, assembly-line style operation with 13 special-purpose machines that, together with his custom-made lathe, produced the stocks without the need for skilled labor.

Now, Clarke (1992: 17) argued that there 'was nothing original in either the detail or the general principles' of Ford's system, only that he applied them to a new industry and did so with such 'single-minded ruthlessness that he transformed the condition of production of motor vehicles almost overnight'. It is true that Singer produced half a million sewing machines in 1880, as many cars as Ford produced in 1916. But there are two critical differences. First, the Model T had more than 10,000 parts – orders of magnitude more complex than the sewing machine. Second, capitalist firms before Ford had no plans to produce low-cost products for a mass market. It was Ford's focus on the mass market that was the real game changer.

A final issue concerns whether Ford's system was in fact widely replicated throughout the US economy after the 1920s. The short answer is that there were significant modifications made to Ford's Model T system, but that these are best interpreted not as the demise of fordism but as the refinement and generalization of the fordist model of mass production. Clarke (1992) argued, correctly, that Ford's Model T system failed by the end of the 1930s as his company became unable to maintain high wage costs (within the existing labor-market regime of wage-based competition), was forced to diversify products under competitive pressure from General Motors, and forced to concede unionization in 1941. At the same time, Williams and collaborators showed that the Ford Highland Park plant in the early twentieth century did not fit the stereotype of fordism, because it included flow production rather than batch production. Taken together, these views are both consistent with my interpretation: the Model T system was of such an extreme level of single-purpose orientation that it ultimately failed and had to be replaced by a modified system – which has widely come to be known as fordism.

Williams et al. (1992: 529–30) persuasively demonstrated that Highland Park was

'proto-Japanese', having anticipated modern lean practices: flow production with low buffer stocks and continuous improvement in reducing direct labor and improving workflow via plant reorganization. However, they are unpersuasive in their claim that inflexible, 'dedicated equipment … cannot be found at Highland Park'. In their interpretation, Ford's machinery 'was flexible in the sense that it was mainly reusable … most of Ford's metal working was done on bought-in, series built lathes, drills, millers and presses'. They conclude that 'special jigs and fixtures' were easily retooled for new products. They cite a single source suggesting Ford's machine tools were easily retoolable, while Hounshell references hundreds of sources throughout his book showing the opposite. To begin with, it must be remembered that the Highland Park plant built a single product, the Model T, and that the changeover to the Model A took more than six months. This was largely because of the single-purpose nature of Ford's machines: of the 43,000 machine tools at the factory, more than half were scrapped, and others refurbished, for the Model A.

It is true that Ford had a tool and die shop of skilled machinists using general purpose lathes, mills, planers, drills and grinders, but these were largely used to build and maintain single-purpose machine tools. The latter were truly single-purpose machines; in the engine department alone such machines included, according to a journalist from *American Machinist*:

> special block and head spotting machines … special machines [that] bored out the cylinders and the combustion chambers on the head … another machine tool drilled at one time forty-five holes in four sides of the block [using] non-adjustable spindles … a drilling machine for babbitt bearing anchor holes … broaching machines for valve stem bushings. (Hounshell 1984: 233)

These were not machines that could be retooled. Other machines may have been of a more general-purpose type, but these were fitted with hard, expensive, often very large single-purpose fixtures, tools and dies.

Ford's system of single-purpose machinery dedicated to making a single product was never going to have a long life. It was simply not feasible to have such a high level of dedicated, extremely expensive capital equipment in an ever-changing market economy. The cost of retooling and reconstruction for the Model A was around $18 million (Hounshell 1984). When making annual model changes – and later, variations on the same model – flow production with dedicated machinery was impossible. There are two issues here. First, in terms of technology, while special-purpose machines undoubtedly continued to be used in mass production, there was a growing use of general-purpose machine tools. This shift was famously implemented at GM to support its annual model-change strategy, under William Kundsen in 1927 (Hounshell 1984). In addition, as Sabel (1982) has argued, to a certain extent whether a machine tool is special purpose or general purpose depends on the know-how of engineers and the skill of operators.

Second, in terms of workflow, the use of more general-purpose machines to produce more varied products made flow production extremely difficult – a problem not fully solved until the development of the Toyota production system. Knudsen thus introduced a decentralized but still vertically integrated structure – Chandler's (1962) multidivisional form – in which there was functional differentiation between plants owned by GM but autonomously run by local management: motors and body stampings in Flint, forgings and axles in Detroit, transmissions in Toledo, carburetors in Bay City, and seven assembly plants located in large cities. This development initiated a shift from the flow production system of the Model T to a batch-and-queue production system. Even inside less decentralized factories, with significantly more parts running through an integrated factory, it was necessary to develop buffer stocks between various operations, which ensured continuity of production (Tolliday and Zeitlin 1992b).

The Classical Fordist Production Regime

Based on the foregoing, Table 16.1 presents a periodization of phases of capitalist development in the US. First *American system* is dated from Blanchard's introduction of a sequence of single-purpose machines to produce interchangeable gunstocks in 1826. This system was based on the use of standardized parts and products, single-purpose machines and mid-volume batch production using semi-skilled labor. The American system also used internal contracting, in which the capitalist provided a factory, machines, power and raw materials, while a contractor brought his own workers into the factory and was responsible for all aspects of labor management. The contractor was paid piece-wages, but his workers were paid time-wages (Littler 1982: Chapter 7). This system was used throughout the New England arms factories and at McCormick (mechanical reapers) and Pope Manufacturing. Singer was among the first to eliminate the internal contractor system, switching in the 1880s to a system of direct hires and foremen, in order to have increased control and better information on production costs. The decline of the internal contracting system led to the rise of piece-wages for individual workers, generating new problems as the latter, fearful of the threat of rate cutting, responded by 'soldiering', that is, systematically restricting output.

Second, *Ford's Model T system* (Table 16.1), is best conceived as a transition period to the stage of fully fledged classical fordism. It is dated from 1913 to 1927, from the beginning of the introduction of the assembly line to the final year of production of the Model T. As noted, the Model T system continued the new practice of producing standardized parts and a single, standardized product with special purpose machinery. Although Ford never explicitly adopted taylorism, he certainly employed semi-skilled workers along principles similar to it. Indeed, as with fordism, taylorism is best thought of not as a system conceived and perfected by an individual (Frederick W. Taylor), but a system that was part of a broader movement of rationalization, in which many individuals made contributions (e.g. Gantt, Barth, Cooke, the Gilbreths, Bedaux), and for which Taylor was the most influential proselytizer. While the pure version of Taylor's system, like that

Table 16.1 Core organizational models, USA, 1826–present

	American system, 1826–1913	*Ford Model T system, 1913–1927*	*Classical Fordism, 1927–1970s*	*Postfordism 1970s–present*
Products/ industries	- Arms - Sewing machines - Typewriters - Bicycles	- The Model T	- Durable goods - Nondurable goods - Fast food	- Manufacturing - Retail sales - Leisure and hospitality - Healthcare - Banking and insurance - Public sector
Characteristics	1. Standardized parts and products 2. Single-purpose machines 3. Mid-volume, batch production 4. Semi-skilled labor 5. Market-determined wages 6. Internal contracting	1. Standardized parts and products 2. Single-purpose machines 3. High-volume, flow production 4. Semi-skilled labor 5. High day wage 6. Vertical integration/ internal labor markets	1. Standardized parts, diversified products 2. General-purpose machines 3. Forecast-driven, high-volume, batch production 4. Taylorism 5. Collectively bargained and administratively determined wages 6. Vertical integration/ internal labor markets	1. Standardized parts and scripts, diversified products 2. Flexible machines 3. Demand-driven, high-volume, flow production 4. Neotaylorism 5. Market-determined wages 6. Vertical disintegration

Source: Author's analysis.

of Ford's, was not widely adopted, many of his disciples and competitors did successfully diffuse a system of task fragmentation, process standardization and sharp separation of conception from execution.

Ford's attempt to solve the problem of soldiering, of course, was the five-dollar day wage. Lazonick (1983) showed that this high wage worked in reducing labor turnover and increasing productivity – but only for a short period. As productivity increases began to slow and competition began to catch up in the 1930s, Ford was no longer able to afford such a high wage. Close supervision could not ensure reliable and attentive workers either. The solution was found in providing internal promotion opportunities within the firm. Internal promotion ladders, and wages associated with positions rather than individuals, are part of a broader set of bureaucratic employment practices known as internal labor markets. Ford and US Steel were pioneers in this area.

As Jacoby (1984) demonstrated, however, internal labor markets were resisted by most employers and were implemented only under pressure from unions, the personnel management movement, and the state. Before the First World War, employment was unstable and inequitable. The first wave of internal labor markets was implemented following the War under pressure from skilled worker unions and reformers (social workers, academics, ministers, vocational guidance counselors), who sought to replace the arbitrary authority of the foreman with personnel departments. A minority of manufacturers began conceding these demands due to labor shortages and labor unrest during the War. The Federal government provided training for personnel managers at universities, but it was not until after the Great Depression that a majority of manufacturing firms implemented personnel departments. Specifically, the National Industrial Recovery Act of 1933 rejuvenated the labor movement, leading non-union firms to adopt personnel departments in an attempt to stave off unions. Promotion ladders did not become widely adopted until the 1940s, under the influence of government agencies including the National Labor Relations Board, the War Manpower Commission and the War Labor Board.

Third, classical fordism (Table 16.1) consists of a refinement of Ford's Model T system. Like the latter it is based on the high-volume production of standardized parts for mass markets. GM introduced two critical adaptations with its strategy of more diversified, but still highly standardized products: general-purpose machines and decentralized, batch production. The strategy of diversified products and the structure of batch production both required improvements in forecasting, leading to the establishment of the classical model of forecast-driven, high-volume, batch production. This fordist model was increasingly associated with taylorist work organization. In keeping with the attempt to substitute management for the market (Chandler 1962), the fordist corporation implemented internal labor markets, including the payment of administratively determined wages associated with positions on an internal promotion ladder, rather than with individual performance (Edwards 1979; Osterman 1984). While the fordist core was located primarily in the manufacturing sector, fordist principles – high volumes, standardization and taylorism – were adopted in leading service companies, most prominently by McDonalds and the emerging fast-food industry (Leidner 1993).

Finally, I briefly discuss *postfordism* (Table 16.1) based on my analysis of the US case (see Vidal 2012, 2013a).[1] The postfordist regime continues the fordist strategy of producing standardized parts and diversified products. I have added standardized scripts to the list given the importance of interactive service labor in the postfordist economy, heavily concentrated in retail sales and leisure and hospitality (Vidal 2013a), much of which is scripted in an attempt to standardize service provision (Leidner 1993). All of the other characteristics of the post-fordist growth regime have been significantly modified or fully reversed from their fordist forms.

Beginning in the 1970s and '80s, the diffusion of microprocessors, CNC machine tools, programmable stamping presses with soft tooling, and quick changeover methods developed in Japan, a new generation of flexible machine tools surpassed the previous generation of general-purpose machines. Production is still high-volume, but, following the path-breaking innovations of Toyota, the model of lean production has now become dominant throughout the manufacturing world (Vidal 2011): demand-driven rather than forecast-driven production, and a return to flow principles based on just-in-time inventory and production control principles. Under the lean model, process standardization is central but it now involves workers in problem solving and is hence neotaylorist (Adler 1995). Similar principles – customer focus, flow production, neotaylorist – are being widely implemented in the service sector, with the healthcare, banking and insurance, and civil service sectors leading the way (Vidal 2011). Finally, there has been a dramatic move toward vertical disintegration, which has largely eliminated internal labor markets and has been implemented along with a return to the market determination of wages (Vidal 2013a).

THE RELUCTANT FORDIST PRODUCTION MODEL IN THE UK

The Industrial Revolution and the Precursors to Reluctant Fordism

Britain, of course, is the birthplace of the Industrial Revolution, beginning around the 1760s and running through the 1830s, including new tools, fertilizers and harvesting techniques in agriculture, the mechanization of textile production (the water frame for the cotton spinning wheel, the spinning mule and the power loom), the steam engine powered by refined coal, and advances in iron making (the use of coke and the rolling mill). Goods production in pre-industrial society was done on a small scale either by households in the cottage/domestic industries or by skilled handicraft trades under the medieval, urban guild system. According to Marx (1990 [1867]), a gradual and uneven transformation took place from the mid-sixteenth to the last third of the eighteenth century in which capitalists combined the distinct trades together necessary to produce a final complex product, such as a carriage, or combined the same trades together in order to improve the division of labor within the trade. Over two centuries machines were introduced into manufacturing and, beginning in the nineteenth century, the factory system began to replace the artisanal system of putting-out to households or small workshops.

In terms of product and process standardization, Britain was very late to the game compared with the US and Germany. With its empire markets, the UK had the largest product market in the world during the nineteenth century (Gospel 2014). However, these markets were highly fragmented and heterogeneous, divided along class and regional lines. In tailoring to these diverse markets, British firms made a wide range of products, with little incentive to standardize products or develop mass production methods (Gospel 1992). Yet, when compared with the early rationalization movements of the US and Germany, explaining the persistence of non-standardized, small-batch craft production well into the 1920s requires further explanation.

The lack of standardization of products and processes was in part a contingent outcome of class struggles. Within management, engineers in the UK lost out in a struggle for control with accountants, because the former failed to establish control over a broad range of activities or to deal with short-term financial problems, whereas the latter succeeded (Ackroyd and Lawrenson 1996). Because British engineers failed to gain a strategic role in management, as they had in the US and Germany, their priorities were subordinated to those of accountants, leading to a failure of overall strategic management.

In the class struggle between management and labor, the former failed to gain sufficient control over the production process (Lewchuk 1983). The informal shop steward movement, which was independent of the national union movement, had gained substantial control over the shop-floor. As a result, management saw the problem of labor cooperation and 'the human element' as the main issue. They prioritized motivation over machine pacing and direct control; the primary mechanism for motivation was the piece-wage and then the premium bonus system, and machine pacing was seen as largely inconsistent with their motivation strategy.

Craft control was more easily eliminated in the US because capital was more mobile and able to relocate to non-union areas (Lazonick 1983), where mechanization and taylorism could be implemented without resistance from craft workers and their unions. Unskilled and semi-skilled labor in the US was not organized until the 1930s, and this large pool of workers included a large immigrant contingent, creating a wide gulf between employer and employee, facilitating the dehumanization of labor (Littler, 1982). In the UK, labor was skilled, organized and more ethnically homogeneous, leading employers to have more concern for the conditions of workers and making their primary concern the threat to their control from organized labor.

The Reluctant Fordist Production Regime

A large firm sector began to emerge before the First World War and industrial concentration rapidly increased during the 1920s. Oligopolies were established in chemicals, electrical engineering, food, drink and tobacco (Gospel 1992). During the War the government encouraged standardization of parts and products, there was an increasing use of mechanization and semi-skilled workers, and the stable markets created by the War gave further impetus to the development of mass production methods (Gospel 1992; Littler 1982). The Ministry of Munitions intervened during wartime production 'by discouraging product differentiation and by encouraging investment in long runs and mass production to meet government demand' (Hannah 1983: 30). There was an intense merger wave in the 1920s leading to a substantial increase in industry concentration, and there were moves toward vertical integration in the food and beverage industries, with the rise of giant firms including Unilever, United Dairies, Boots and Cadbury-Fry. During this period the professionalization of management grew. The form of taylorism that finally gained traction in the UK was known as the Bedaux system, after Charles E. Bedaux (Littler 1982). He developed a form of taylorism that was said to be simpler to install and required less encroachment on the authority of traditional management.

During the Great Depression, Britain turned to protectionism and collusion, which greatly facilitated vertical integration and the introduction of internal labor markets (Gospel 2014). After the Second World War concentration increased dramatically. Within manufacturing, the proportion of employment in giant firms (those with more than 1,500 employees) rose from 15 percent in 1935 to 24 percent in 1951, strongly suggesting a transition to mass production techniques. By 1957 the 100 largest firms held 60 percent of net assets in the manufacturing sector and by 1969, 75 percent. The number of UK firms producing TVs declined from 60 in 1954 to just seven in 1969.

In sum, the production model that Britain achieved by the postwar period may be referred to as *reluctant fordism*. This model is compared with classical fordism in Table 16.2. British manufacturing had finally become organizationally fordist, including the use of standardized parts, general-purpose machines, forecast-driven, high-volume production. I refer to the system as *reluctant* fordism because fordist ideas were never fully accepted

Table 16.2 Fordist regimes, USA, UK, Germany, 1945–1973

	USA	*UK*	*Germany*
	Classical Fordism	Reluctant Fordism	Flexible Fordism
Core production model	1. Standardized parts 2. General-purpose machines* 3. Forecast-driven, high-volume, batch production 4. Classical taylorism 5. Collectively bargained and administratively determined wages 6. High vertical integration and arms-length interfirm relations	1. Standardized parts 2. General-purpose machines 3. Forecast-driven, high-volume, batch production 4. *Modified taylorism (Bedaux)* 5. Collectively bargained and administratively determined wages 6. *Moderate vertical integration and arms-length interfirm relations*	1. Standardized parts 2. General-purpose machines 3. Forecast-driven, high-volume, batch production 4. *Democratic taylorism* 5. Collectively bargained and administratively determined wages 6. *High vertical integration and interfirm cooperation*
	Liberal Fordism	Blocked Fordism	Nonliberal Fordism
National growth regime	1. Core employment sector: oligopolistic manufacturing 2. Mass consumption via mass domestic market 3. Supply-driven, producer-dominated supply chains 4. Liberal Keynesian welfare state 5. Class compromise: national union contacts and pattern bargaining 6. Well-integrated financial and productive capital investment banks	1. Core employment sector: oligopolistic manufacturing 2. Mass consumption via domestic and European Common Market 3. Supply-driven, producer-dominated supply chains 4. Liberal Keynesian welfare state 5. *No class compromise due to obstructive working-class politics* 6. *Unintegrated financial and productive capital due to fractionalized capital*	1. Core employment sector: oligopolistic manufacturing 2. Mass consumption via domestic and European Common Market 3. Supply-driven, producer-dominated supply chains 4. Corporatist Keynesian welfare state 5. Class compromise: centralized, national-level pattern bargaining 6. *Deeply integrated, long-term relations between financial and productive capital via* Hausbanken

Note: *I follow Hounshell (1984: 265–6) in distinguishing the single-purpose machine tools used from the 1850s to the 1920s in the US, from the subsequent generation of more general-purpose machine tools introduced at GM in the 1920s. The latter, which were more standardized and hence allowed a quicker changeover, can be differentiated from flexible (programmable) machine tools and presses that use microprocessors, which diffused in the 1970s and '80s.

Source: Author's analysis.

by a large proportion of British management. This reluctance can be seen in two partial but important modifications to the classical model. First, a modified form of taylorism, the Bedaux system, was adopted, and skilled workers continued to remain more important to production than in the US. Second, the UK remained less vertically integrated than the US and more dependent on arms-length relations between firms. Still, during the post-war period, internal labor markets in core firms were formalized and structured, with rules and practices for internal promotion by seniority, and wages associated with positions in the hierarchy, although there continued to be underinvestment in firm-specific training (Gospel 1992). Britain achieved high wages through class struggle in which structural conditions made labor more powerful than capital, not through a class compromise as in the US and Germany. There were national-level agreements but significant wage drift due to local bargaining (with strong stewards and weak managers) in a context of full employment and strong demand.

FLEXIBLE FORDISM IN GERMANY

Precursors to Flexible Fordism

The story of German industrialization is much less contested than that of the UK. This section

will thus be much briefer than the previous sections. 'The consensus seems to be', according to Kocka (1980), that Germany's first phase of *large-scale* industrialization 'began about 1840 and ended with the economic crisis of 1873'. Germany followed a similar trajectory to the US: the construction of railroads, and other traditional industries, creating a large integrated domestic market. In turn, the German economy developed a core of manufacturing firms that were largely fordist, although in certain respects more flexible than their American counterparts. Now, Herrigel (1996: 18) shook up the consensus, making the critical point that although fordism was dominant in some regions, other regions were dominated by dynamic economies of decentralized, specialized small and mid-sized firms and craft producers. The latter, derived from pre-industrial roots in the seventeenth and eighteenth centuries, experienced their own independent form of industrialization, and flourishing alongside the fordist regions well into the postwar period. It is thus more accurate to say large-scale industrialization emerged, in Herrigel's words, 'alongside an already existing and still robust system' of craft producers. This form of industrial dualism is fully consistent with my regulationist theory of combined and uneven development; such heterogeneity exists within and as across national economies (Vidal 2014).

The rise of the vertically integrated, diversified firm emerged from the 1870s through the First World War (Chandler and Daems 1980; Kocka 1980). In some respects, German moves toward fordist-type organization occurred earlier than in the US. Most importantly, as early as 1887, 14 of the top 100 largest firms were fully integrated (forward into distribution and backward into supply), and by 1907, 62 of the top 100 were (Kocka 1980). These firms were also highly diversified, more so than US firms at the time (Chandler and Daems 1980). Cartelization came along with vertical integration and diversification. Since laissez-faire ideas had never become strongly accepted in Germany, whose business leaders were more concerned with stability, cartels were widely seen as legitimate forms of business organization (Djelic 1998; Kocka 1980). By 1907, cartels accounted for around 25 percent of total industrial output, and by 1930 there were around 2,100 cartels. German industry thus became heavily oligopolistic. In 1907 Siemens and AEG together employed around two-thirds of the electrical engineering industry, and still 40 percent of the workforce by 1939.

The first industrial boom in Germany, around the 1850s, was financed by banks. Germany was similar to America and different from Britain insofar as it had a very close relationship between finance and industry. However, the nature of that close relationship differed radically, with repercussions that continue to this day: America had a highly developed capital market whereas Germany did not. German banks developed a unique system of universal banking modeled on a combination of British commercial banks and French investment banks, allowing them to provide investments to industry funded by short-term deposits from commercial banking (Gerschenkron 1962). Beginning in the 1890s, banks and industry began to develop very close relations (Kocka 1980). After this period banks shifted from providing short- and long-term loans to providing primarily long-term loans. Having long-term stakes in business, banks increasingly moved to secure direct influence in firm strategy.

In Germany the development of professional and bureaucratic management took place around the same time as the US and earlier than the UK, but the complete separation of ownership from control occurred later than both (Kocka 1980). The emphasis on technical training and engineering was strong in Germany, and between 1890 and 1930 it surpassed even the US in the proportion of top managers – owners and salaried – with some academic training. In the context of a strong bureaucratic tradition from the history of German absolutism (Homburg 1983), which emphasized rules, procedures and standardization, it is not surprising that Germany

was an early adopter of product standardization and professional managerial hierarchies. Indeed, large German firms developed ideas on systematic factory organization similar to those of Taylor without having been exposed to his ideas (Kocka 1980). Upon hearing of taylorism, German managers and engineers eagerly visited the US to study it, and by 1907 it was widely discussed in Germany. After the First World War, the German electrical engineering firms, including Siemens, helped to organize associations to promote taylorism, including the National Efficiency Board, the National Committee on Time Studies, and the National Board for Industrial Standards (Homburg 1983).

The Flexible Fordist Production Model

Herrigel (1996: 150) notes that during the interwar period mass production was introduced in the steel, auto, light machinery and textile industries. The producer goods industries, including machine tool makers, were also large, bureaucratic firms, but they produced more customized products in smaller batches, using general-purpose machinery and skilled labor. As early as the First World War, Daimler and Benz were adopting methods more along the lines of classical fordism than the Model T system: a mix of dedicated and general-purpose tools, volume production in large batches based on functional (rather than flow) organization and increased vertical integration. The German batch producers were low-volume, custom producers using skilled and largely autonomous labor within a tradition of labor-management cooperation. Over the interwar period, however, the small-batch, custom machinery producers began to produce standard machines, and along with this increased standardization and volumes.

The consolidation of the fordist model occurred rapidly after the Second World War. 'The American system of industrial production', argues Djelic (1998: 2) 'was constructed after 1945 as a universal model for the Western world'. Institutional mechanisms for transferring the fordist model to Western Europe were established by American involvement in reconstruction, most notably through the Marshall Plan – administered under the newly created Economic Cooperation Administration (ECA) in 1948 – and the American Military Government in Germany (OMGUS). East Germany, of course, was under Soviet control. The ECA and OMGUS explicitly adopted fordism as their model, understood as 'large, mass producing corporations [competing] on oligopolistic markets' (Djelic 1998: 104). Over the 1950s 'the Ministry for the Marshall Plan published close to one hundred reports on every aspect of the American system of industrial production – mass production, standardization, specialization, rationalization, antitrust, human relations, trade unions' (Djelic 1998: 181). Five thousand Germans were sent to observe American production techniques between 1950 and 1951.

In sum, in the postwar period the dominant production model in Germany came to be *flexible fordism* (Table 16.2). Like the US and the UK, German manufacturing widely adopted standardized parts, general-purpose machinery and forecast-driven, high-volume batch production. The German system achieved more flexibility through a system of democratic taylorism (a term Adler (1995) has used to describe Japanese lean production), based in a Christian and social democratic tradition of labor-management cooperation, which was institutionalized in the Works Councils Act of 1952, providing for local works councils and worker representation on supervisory boards. While German manufacturing was also highly vertically integrated, high levels of interfirm cooperation between lead firms and their suppliers provided an additional source of flexibility. Finally, based in the Bremen Agreement of 1956, national, centralized bargaining between the metalworkers union (*IG Metall*) and the Metal Industry Employers Association (*Gesamtmetall*), was used as a benchmark for the patterning of national wages in industry (Herrigel 1996: 265).

CONSOLIDATING AN INTERNATIONAL GROWTH REGIME: ATLANTIC FORDISM AND THE GOLDEN AGE

Before a brief comparison of the national growth regimes in the US, UK and Germany, it is first necessary to discuss the Atlantic fordist international growth regime, which established the structural basis for the national regimes. As we have seen, industrialization is a combined and uneven process that takes place initially at the regional level. Yet, the dynamics of capitalist development are inherently international, indeed global, insofar as the tendency of capitalist production is toward unlimited growth, which only respects national boundaries when forced to. International institutions were critical to the facilitation and consolidation of fordist development. First, the Bretton Woods agreement established an international monetary regime which allowed national policy autonomy and hence the establishment of Keynesian welfare states. Second, the Common Market established by the European Economic Community provided a basis for mass markets.

The Bretton Woods system and the Common Market undergirded a fordist dynamic of domestic mass production and mass consumption in the US and Western Europe. The European countries that most fully realized this regime were Britain, Germany and France. As Jessop (2002: 58) has argued, the smaller open economies (Austria, Denmark, Sweden, Canada) realized a mass consumption society 'because they occupied key niches in an international division of labor whose transatlantic dynamic was decisively shaped by the leading fordist sectors in the leading economies'. Even where the smaller countries were primarily producers of small batch capital or luxury goods, they were able to share in the growth dynamic of Atlantic fordism, consuming mass produced goods via income from export demand.

The most important element of the Atlantic fordist regime was the Bretton Woods agreement of 1944, in which the major industrialized economies agreed on a system of capital controls considered necessary to protect the policy autonomy of Keynesian welfare states 'from being undermined by speculative and disequilibrating international capital flows' (Helleiner 1994: 4). In the open system of international finance during the late nineteenth and early twentieth centuries, speculative flows caused severe volatility in exchange rates and trade relations, widely being seen as having been a central cause of the Great Depression. After 1931, states turned to protectionism in trade and capital controls in attempt to stabilize their economies. As Helleiner shows in great detail, the Bretton Woods agreement was made possible because, following the Great Depression, private and central bankers had lost influence and been displaced from positions of power by a coalition of Keynesian-oriented state policymakers, industrialists and labor leaders.

The Atlantic fordist regime – including Keynesian welfare states and economies of fordist mass production/mass consumption in the largest economies – produced what has been widely referred to as the Golden Age of (Atlantic) capitalism, conventionally dated from 1950 to 1973, when Bretton Woods collapsed and a series of macro indices began to turn. As shown in Table 16.3, GDP growth rates in the top five OECD economies during that period averaged a remarkable 5.4 percent, a level not seen before or after. Similarly, Table 16.4 shows labor productivity for the same states and period averaging 4.9 percent, with rates before and after rarely approaching even half of that. After 1973, in

Table 16.3 Phases of GDP growth, 1870–1984*

	1870–1913	*1913–50*	*1950–73*	*1973–84*
US	4.2	2.8	3.7	2.3
UK	1.9	1.3	3.0	1.1
Germany	2.8	1.3	5.9	1.7
France	1.7	1.1	5.1	2.2
Japan	2.5	2.2	9.4	3.8
Average	2.6	1.7	5.42	2.2

Note: * Average annual compound growth rates.

Source: Maddison (1987).

Table 16.4 Phases of growth in labor productivity, 1870–1984*

	1870–1913	*1913–50*	*1950–73*	*1973–84*
US	2.0	2.4	2.5	1.0
UK	1.2	1.6	3.2	2.4
Germany	1.9	1.0	6.0	3.0
France	1.7	2.0	5.1	3.4
Japan	1.8	1.7	7.7	3.2
Average	1.7	1.7	4.9	2.6

Notes: *GDP per hour worked; average annual compound growth rates.

Source: Maddison (1987).

addition to marked declines in output, productivity, profit rates and export growth, the OECD also experienced rising unemployment, inflation and export and GDP instability (Glyn et al. 2007). Finally, although the historical data on income inequality are patchy, existing data suggest that within the states of the Atlantic fordist regime there were rising or stable inequality levels up to the Great Depression and declining inequality during the 1930s, which was maintained throughout the fordist period. Following the crisis of fordism in the 1970s, inequality rose in many of these countries. According to Piketty's (2014: Figure 9.8) dataset on the share of income going to the top 10 percent of the population, inequality in the US rose from 1900 to around 1930, then declined steadily until around 1973, and has risen ever since. In Europe (UK, Germany, France, Sweden) inequality remained stable from 1900 to 1910, declined over the 1910s, remained stable again for the 1920s, and then declined dramatically from 1930 to 1980, after which it rose to the present level.

Since the collapse of the Bretton Woods system in 1973 – and the long, slow transition from national fordist institutions – most of the countries in the Atlantic fordist regime have experienced stagnant growth, declining or stagnant profit rates, and rising inequality (Glyn et al. 2007; Vidal 2013b). By the early 1970s unilateral capital controls were unable to contain speculative flows. Following a massive round of capital flight from the US to Europe in 1971, the former suspended gold convertibility, forcing Europe and Japan to temporarily float their currencies. The Europeans tightened their capital controls in an attempt to stop the revaluation of their currencies, but they failed, and in 1973 they permanently floated their exchange rates, signaling the end of the Bretton Woods system.

A restrictive financial order was created under Bretton Woods in order to facilitate trade liberalization (Helleiner 1994). This concern dovetailed with the concerns of the Marshall Planners, who were fully aware that the model of fordist mass production they were using to reconstruct Europe could be established only if there were a mass market (Djelic 1998). Accordingly, they advocated union in Europe. The Organization for European Economic Cooperation (replaced by the OECD) was created in 1948, the European Payments Union (replaced by the European Monetary Agreement) in 1950, the European Coal and Steel Community (forerunner to the European Union) in 1951, and the European Economic Community or Common Market (enlarged and transformed into the EU) in 1957. As a result of these international associations and agreements, tariffs were removed within much of Western Europe, creating a large internal European mass market similar to the US (Maddison 1987).

CONSOLIDATING THE NATIONAL GROWTH REGIMES: AMERICAN LIBERAL, BRITISH BLOCKED AND GERMAN NONLIBERAL FORDISM

The comparative analysis is summarized in Table 16.2 (bottom panel). The American growth regime is referred to here as *liberal fordism*, due to the fact that the postwar institutional settlement included a liberal welfare state and highly marketized relations between finance and industry. Following Peck and Tickell (2000), the British growth regime is referred to as *blocked fordism* due to a failure

to realize a national class compromise between capital and labor, and a lack of integration between finance and industry. The German growth regime is referred to as *nonliberal fordism* due to its quasi-corporatist welfare state and industrial relations institutions, and to its deeply integrated, long-term relations between finance and industrial capital.

The bottom panel of Table 16.2 lists six characteristics of each national growth regime, which I discuss in order. First, the core sector of employment in all three countries in the 1950s and 60s was manufacturing. Table 16.5 shows the aggregate statistics for each country along with France and Japan for comparison. Based on this, one rough measure of fordism (or mature industrialism) might be when manufacturing accounts for one-third or more of employment; and for postfordism (or postindustrialism) when services account for 60 percent or more of employment. On this measure all four countries except Japan were fordist by 1950 and all but Japan and Germany postfordist by 1984.

The manufacturing sector in each country was also oligopolistic. Within the car industry, according to Tolliday and Zeitlin (1992b: 6), 'Until the late 1960s each national market in Europe remained quite distinct and dominated by one or two domestically produced cheap small cars'. More broadly, in the US, the largest 100 firms accounted for 22 percen of net manufacturing output by 1919, 30 percent by 1954 and 33 percent by 1963. In the UK, the largest 100 firms accounted for 22 percent of net manufacturing output by 1924, 27 percent by 1953 and 41 percent by 1968 (Prais 1981). In Germany, the four-firm concentration ratios in 1958 were 61 percent for petroleum refining, 82 percent for cars, 49 percent for steel (in 1960), 98 percent for roller bearings, 71 percent for refrigerators and freezers, 66 percent for glass bottles and 86 percent for cigarettes (Müller 1976). By the end of the fordist period, industrial concentration in the UK had surpassed that in the US and Germany, with 90 percent of net output accounted for by just five firms in

Table 16.5 Employment structure, 1870–1984*

		US	*UK*	*Germany*	*France*	*Japan*
Agriculture	1870	50.0	22.7	49.5	49.2	67.5
	1913	32.2	11.0	34.6	37.4	64.3
	1950	13.0	5.1	22.2	28.5	48.3
	1960	8.2	4.6	13.8	21.9	30.2
	1973	4.1	2.9	7.2	11.0	13.4
	1984	3.3	2.6	5.5	7.6	8.9
Industry	1870	24.4	42.3	28.7	27.8	13.8
	1913	29.3	44.8	37.8	33.8	13.9
	1950	33.3	46.5	43.0	34.8	22.6
	1960	34.3	46.7	48.2	36.4	28.5
	1973	32.5	41.8	46.6	38.4	37.2
	1984	28.0	32.4	40.5	32.0	34.5
Services	1870	25.6	35.0	21.8	23.0	18.7
	1913	38.4	44.2	27.6	28.8	21.8
	1950	53.7	48.4	34.8	36.7	29.1
	1960	57.5	48.7	38.0	41.8	41.3
	1973	63.4	55.3	46.2	50.6	49.4
	1984	68.7	65.0	54.0	60.4	56.3

Note: *Percentage of total employment.

Source: Maddison (1987).

one quarter of all manufacturing industries (Gospel 1992). Davis and Cobb (2010) demonstrated that industry concentration ratios have a remarkably strong negative correlation within inequality, with a correlation of –0.8 in the US from 1950 to 2006. Across a range of other European, Asian and Latin American countries increased industry concentration was associated with decreased inequality. Inequality is reduced by high concentration due to the substitution of administrative policies regarding wages and promotions for market determination.

Second, all three growth regimes were based on mass production and mass consumption, which was achieved via a mass domestic market in the US and, in Western Europe, a combination of domestic markets and the European Common Market. As shown in Table 16.6, the US was effectively a self-contained economy, with total imports accounting for only 2.9 percent of GDP in 1962, and exports just 3.8 percent. If we consider the European Common Market as a single economic region then it was nearly as closed as the US economy, and certainly far less open than might be thought, given the common characterization of Germany and the UK as export-led economies and Northern Europe as comprising a number of small, open economies. While imports amounted to 16.2 percent of European GDP in 1962, 4.4 percent of that was from non-OECD countries, primarily raw materials, and of the remaining 11.76 percent that came from the OECD, fully 76 percent came from other European countries. Thus, considering Europe as a single economic region, imports as a percentage of GDP in 1962 amounted to just 7.2 percent versus 2.9 percent for the US. Exports accounted for just 3.8 percent of American GDP in 1962 and (excluding intra-Europe trade) 5.5 percent in Europe. In short, during the fordist period Europe had established a virtuous circle of mass production and mass consumption in much the same way as the US. According to Tolliday and Zeitlin (1992b), up until the late 1960s, trade between countries was complementary,

Table 16.6 Imports and exports of goods as a percentage of nominal GDP

		Imports	
Importing country/ region	*Source*	*1962*	*1972*
United States			
	OECD	1.73	3.34
	Non-OECD	1.13	1.26
Europe*			
	OECD	11.76	14.13
	Europe	8.93	11.71
	Rest of world	2.83	2.42
	Non-OECD	4.40	3.94
		Exports	
Exporting country/ region	*Destination*	*1962*	*1972*
United States			
	OECD	2.37	2.77
	Non-OECD	1.42	1.34
Europe			
	OECD	10.61	13.74
	Europe	8.80	11.69
	Rest of world	1.81	2.05
	Non-OECD	3.64	3.44

Note: *Europe: Austria, Belgium, Denmark, Finland, France, Germany, Greece, Iceland, Ireland, Italy, Luxembourg, Netherlands, Norway, Portugal, Spain, Sweden, Switzerland, Turkey, UK.

Source: Organisation for Economic Co-operation and Development (1994).

with little head-to-head trade in competing product lines. Only after domestic/regional markets became saturated at the end of the 1960s did large producers begin to enter into direct international competition with each other, pressured by reduced demand and overcapacity.

Third, producers were the most powerful firms in the economy, as opposed to the post-fordist period in which retailers have come to be the most powerful non-financial firms in the economy by virtue of their control over global supply chains (Vidal 2013a). While there was a high degree of vertical integration during the fordist period, including backward integration into components and, less commonly, forward integration into retailing, it

was still common for such companies buy-in raw materials and subcontract for many components. In these fordist supply chains, giant producers exercised powerful control over both forward and backward linkages (Gereffi 1994). Over time, as fordist producers began to vertically disintegrate and globalize production via complex supply chains, there was a shift in power away from manufacturers to large retailers, who began to exercise control over supply chains and often to dictate the terms of production. The fordist period was characterized by supply-driven, producer-dominated, largely domestic supply chains.

Fourth, Keynesian states in all three countries emphasized the use of fiscal and monetary policy to achieve demand management and full employment. The Keynesian welfare states in the US and UK were heavily liberal (Esping-Andersen 1990; Jessop 2002). Liberal states play only a residual role in providing social welfare (e.g. means-tested assistance and minimal social transfers) and emphasize market solutions to social problems rather than state interventions. Of course, the British National Health Service marks the UK state off from the US with its privatized healthcare system. Nonetheless, liberal states in both countries developed out of, and were institutionally linked with market-regulated economies (Jessop 2002). In contrast, the German Keynesian welfare state is of a conservative-corporatist type, which plays a more interventionist role in the provision of social welfare, although the family and existing status differentials are privileged, with welfare rights attached to status and class; social problems are addressed primarily by the family and the voluntary sector, most importantly the Church, and secondarily by the state. The conservative-corporatist state grew out of and was institutionally linked to a non-liberal economy based on close coordination between industry and finance, and within industry.

Welfare state regimes also powerfully shape the gendering of the labor market, and fordist regimes differed significantly on this dimension (Gottfried 2000). The conservative-corporatist regime in Germany explicitly sought to preserve a traditional model of the family, with men as the sole breadwinners and women as homemakers. Stay-at-home mothers were rewarded with child and family allowances, while the tax system penalized married women who worked full time. Thus high wages, job security and expansive social rights were enjoyed primarily by men. In the liberal regimes of the US and UK, minimal public support for mothers meant that more women had to enter the labor market, which meant women either faced a trade-off between work and family or, because of the broader gender norms in the society, had to work while taking on the primary homemaker role. By contrast, social democratic regimes such as in Sweden, which aimed for a high social wage and low inequality, used universal entitlements, that, by the end of the fordist period, included an array of policies 'ranging from parental leave, subsidized childcare to tax reform, encouraging and enabling women to enter into the labor force by minimizing the trade-off between employment and family formation' (Gottfried 2000: 243). Gottfried characterized the gender regimes under fordism as dual breadwinner with female care in the liberal regime, male breadwinner with female care in the conservative regime, and dual breadwinner with state care in the social democratic regime.

Fifth, the US and Germany realized institutionalized forms of class compromise, while the UK did not. In the US this began with the National Labor Relations Act of 1935 and was fully realized in the Treaty of Detroit, the 1950 agreement between General Motors and the United Auto Workers, in which unions traded labor peace for wages linked to productivity increases and with annual cost-of-living adjustments. The Treaty established a pattern which was widely emulated across industry. In Germany, a class compromise was institutionalized through centralized, national-level bargaining, beginning with the 1956 Bremen Agreement in the metal industry (Herrigel 1996). In both countries the institutionalization of class compromise via collective bargaining dovetailed with oligopolistic product

markets to effectively take wages out of competition and realize a high-wage economy to drive demand.

By contrast, the UK never realized an institutionalized class compromise. Its industrial relations system has commonly been characterized as a voluntarist system, organized entirely by private business associations and unions, with an abstentionist state. Howell (2005) showed that this interpretation is not entirely correct, as the state played a critical administrative role in supporting the development of institutions for industry-level bargaining beginning in the 1890s. Industry-level bargaining was the dominant form until the institutions supporting it began to collapse in the 1950s. The new fordist organizations did not have institutions or mechanisms for internally managing economic change and productivity improvement, leading to intensified workplace-level conflict, unsanctioned and uncontrolled by the national Trades Union Congress. This resulted in a shift to workplace bargaining to establish local mechanisms for negotiating change. Along with rising strikes came wage drift, as local contracts exceeded the industry-level minimums (Gospel 1992; Howell 2005).

Finally, the US and Germany each realized a distinct form of integration of financial and industrial capital, while the UK failed to do so. In the US, industry was financed from the late nineteenth century via large investment banks such as J.P. Morgan & Company and the City Bank of New York. In Germany, we have seen how long-term relations developed via banks and manufacturers; each industrial concern was typically in a long-term, collaborative relationship with a *Hausbank*, a universal bank providing comprehensive services. By contrast, in the nineteenth century, banks in the UK had a commercial rather than industrial orientation. Lash and Urry (1987: 50) argued that because the landed aristocracy became a commercial capitalist class before industrialization, it was focused on international finance. The City of London, Britain's financial center, was engaged more in intermediation between investors and borrowers than long-term investment. Additionally, finance, with its close ties to the aristocracy, had traditionally enjoyed high social status, whereas industry, in particular the new heavy industries, had low status. Another major factor in the underdevelopment of capital markets in early nineteenth century Britain, they suggested, was that firms were 'consistently more secretive' than in the US or Germany, resulting in a major disincentive to invest. Gospel (2014) argued that beginning in the early twentieth century British firms were able to better access equity financing, and Foreman-Peck and Hannah (2012) showed that the separation of ownership from control happened earlier in the UK than the US. Nonetheless, finance and industry remained less integrated in the UK than in the US or Germany over the interwar years, and the long legacy of problematic integration between finance and industry contributed to capital underinvestment in British industry – between 1913 and 1950, average annual compound capital productivity growth rates averaged just 13 percent in the UK versus 56 percent in Germany and 96 percent in the US (Maddison 1987).

CONCLUSION

This chapter has presented a comparative and historical analysis of the industrialization of the UK, US and Germany from the beginning of the nineteenth century through the Golden Age of Atlantic capitalism in the 1950s and 60s. I have sought here to make two basic arguments. First, this history is best understood as the combined and uneven development of regional and national economies within an international space, driven by class struggles over the institutions constituting and regulating the political economy at the organizational, sectoral, national and international levels. Second, the development of fordist institutions in the twentieth century incorporated the working class into capitalism through a mass production/mass consumption

economy, thus producing the Golden Age: two decades of remarkable economic growth and stability driven by middle-class consumption.

At the level of a model of production, following the industrial revolution in the UK, a pre-industrial, artisanal system of cottage/domestic industries and a skilled handicraft trades system began to be replaced in the nineteenth century by a factory system. Even within the British factory system, however, non-standardized, craft production remained dominant. It was in the US in the mid-nineteenth century where standardized parts and processes were developed. Henry Ford built on these innovations to develop a model of high-volume, low-cost production aimed at the mass market. This model was refined and generalized by General Motors, leading to classical fordism, widely adopted in the 1930s: high-volume, forecast-driven, batch production of standardized products in vertically integrated firms using general purpose machines and a taylorist division of labor.

Due to the reluctance of British management, the fordist model was adopted very late there. By the end of the Second World War, the British economy had established a production regime of what I call reluctant fordism, the classical model with two modifications: a modified form of taylorism in which skilled workers continued to retain an important role; and less vertical integration. By contrast, the classical fordist model was widely admired and enthusiastically adopted in Germany. I refer to the production regime there as flexible fordism, since the Germans made some improvements on the classical model: a democratic version of taylorism and high levels of interfirm cooperation between large firms and their suppliers.

At the international level, the Atlantic fordist regime was consolidated by the establishment of the Bretton Woods system (1944–1973), an international monetary regime which allowed national policy autonomy, and the European Economic Community, or Common Market, which provided a basis for mass markets in Europe. At the national level, I labeled the American growth regime *liberal fordism* due to its basis in a liberal welfare state and highly marketized relations between finance and industry. The British growth regime is called *blocked fordism* due to a failure to realize a national class compromise between capital and labor, and a lack of integration between finance and industry. The German growth regime is called *nonliberal fordism* due to its quasi-corporatist welfare state and industrial relations institutions, and to its deeply integrated, long-term relations between finance and industrial capital. Wages were taken out of competition due to a class compromise in the US and Germany, and industry-level bargaining without class compromise in the UK, all of which were rooted in oligopolistic competition and the Bretton Woods financial system.

By the early 1970s the international regime, the national regimes and the organizational regimes were all under severe pressure. At the international level, unilateral capital controls were unable to contain speculative flows and eventually the Bretton Woods system collapsed, leading to the internationalization of finance. This put severe pressure on Keynesian welfare states, leading to the rise of neoliberalism. At the organizational level, fordist production was increasingly rigid in the face of emerging production models and the globalization of production. Under the globalization of finance and production, international, head-to-head competition has increased dramatically, and domestic manufacturing employment has been replaced with service work, all of which has generated a new round of destructive, wage-based competition.

In terms of future research, perhaps the most obvious direction would be to bring Canada, Japan and the remaining European countries into the analytical framework developed here. Concerning the US, the UK and Germany, I have attempted to adjudicate between contending historical interpretations and present a set of realist models of various fordist regimes, but there remain a few issues which continue to be debated or for which there is limited historical evidence. First, there is continued debate over

the whether finance and industry in the UK were deeply disintegrated (Lash and Urry 1987) or whether industry was better able to access finance than has been claimed (Gospel 2014). In particular, questions remain over the nature of relations between industry, capital markets and commercial banks in the UK, how such relations evolved over time, and how much responsibility for the long-term poor performance of the British manufacturing sector can be linked to problematic integration between industry and finance in the early twentieth century? Additionally, more comparative research is needed on the orientation of the financial sectors of the Atlantic capitalist countries – commercial or industrial, domestic or international, etc. – during the early years of industrialization from the late 1800s through the 1930s.

Second, there are disagreements in the literature over the timing and causes of the bureaucratization of labor management. Key issues include the relative balance of power between foremen, professional managers, engineers and accountants (suggestive but incomplete analyses of the UK are found in Ackroyd and Lawrenson (1996) and Armstrong (1984)). The brief evidence presented here suggests that the timing and outcomes of these power struggles differed in each country, with important implications for the developmental trajectory of each, but more research is needed. Additionally, while there is a substantial research literature on the development of internal labor markets in the US (for an overview, see Osterman (1984)), there appears to be less on the UK, Germany and other European countries. One open question concerns the role of institutional actors such as unions, the state and professional associations in the development of internal labor markets across countries.

NOTE

1 In the 1980s and 90s there was a debate in the literature on production models over neo- versus postfordism, where the former has been understood to imply a partial continuation with fordist practice and the latter to imply a complete break with the past. The central point of contention is whether taylorism was replaced with a neotaylorist system of employee involvement in standardizing work, along with work intensification, or a post-taylorist system of self-directed teams of empowered workers, without standardized work (for a review of the debates, see Beynon and Nichols (2006b) and Edgell (2012)). After the dust had settled from those debates, it became clear that those arguing for a complete break had overstated their case, and it was widely acknowledged that there are both continuities (mass production and neotaylorism) and changes (demand-driven production and vertical disintegration) in lean production, the dominant model of postfordism (Vidal 2011). While the neo/post distinction is helpful in terms of taylorism, when applied to the broader concept of fordism it ultimately hinders appreciation of organizational diversity within and across sectors of the economy. Rather than attempting to distinguish production regimes of neo- versus postfordism, it is more helpful to simply use postfordism to refer to the period after fordism, understood to include a dominant but not monolithic production model, in which there are a range of progressive and regressive tendencies, institutional settlements and organizational regimes.

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